Abdoulaye Chaibou Saidou

Engineering student · Internship / research topic — 3D, computer vision, digital twins

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PROFILE

Fifth-year engineering student (IATD-SI, ENSAM Meknès) interested in 3D, computer vision, digital twins and physical AI. I built a 3D augmented-reality visualization prototype (Unity, PyTorch), an equipment monitoring platform on simulated sensors and an image-processing tool, and studied an interpretable vision model.

Fast, continuous learner: picked up Julia, Rust, Unity / AR, RL and MLOps through my projects — I never stop learning.

SKILLS

3D / AR: Unity (C#), augmented reality (placement, touch interaction), real-time rendering
Vision: image processing (Pillow), thresholding, masks, histograms, ProtoPNet (study), data annotation
Simulation / AI: Python, PyTorch (MLP), Gymnasium, Soft Actor-Critic, FFT, Isolation Forest
Systems: FastAPI, WebSocket, Docker Compose, Next.js, Git

EXPERIENCE

Full-Stack Developer Intern — DigiScia, Ouagadougou (Burkina Faso) Jul – Sep 2026

- SimpleOQuotidien project (connecting small-trade service providers with clients: search, quotes, assignment); my part: the front-end, in Next.js.

Data Scientist Intern (remote) — Technocolabs Softwares, Indore (India) Aug – Oct 2025

- Team project “Startup Acquisition Analysis & ML Pipeline”: predicting startup outcomes (acquisition, closure, IPO, continued operation).
- My part: decision trees and SVM on the “companies” dataset.
- Contributed to preprocessing, feature engineering and cross-validation (per the certificate).

PROJECTS

NeurAR — 3D / AR visualization of a neural network Unity 6 (C#), PyTorch, FastAPI, WebSocket

Personal project (prototype)

- Augmented-reality application (Unity, C#) rendering a PyTorch MLP (MNIST) in 3D: neurons and connections layer by layer, animated forward and backward passes.
- AR scene placement, rotation, zoom and touch-based neuron selection.
- FastAPI backend (REST + WebSocket) running the model and streaming activations and weights in real time to the Unity client.

AVIEN — industrial predictive maintenance FastAPI, Celery, Redis, TimescaleDB, Next.js, Docker Compose

Academic project, June 2026 · supervisor Prof. Smail Zaki

- Machine condition monitoring from simulated sensors: FFT vibration diagnostics, anomaly detection (Isolation Forest), health score and RUL estimate, alerts and a Next.js dashboard.
- Limitations stated: simulated data, no quantitative evaluation for lack of labels.

TkImage Studio — image annotation and preprocessing Python, Pillow

Academic project 2025/2026 · supervisor Prof. Brahim Bakkas

- Python desktop application: transforms (crop, rotate, resize), filters, thresholding, masks and ROI, pixel statistics, histogram and intensity map.
- Per-image annotation with JSON / CSV export; image analysis extension through the Gemini API.

ProtoPNet — study of an interpretable prototype network Python, Jupyter

Course project (CUB-200-2011) · teaching work

- Study of the ProtoPNet model (bird classification via prototypes): explanatory notebook and presentation for classmates.
- No contribution to the model or architecture (researchers’ work).

Portefeuille-Advisor — reinforcement learning Gymnasium, stable-baselines3, Streamlit, TensorBoard

Academic project, Jan 2026 · supervisor Prof. Tawfik Masrour

- Gymnasium simulation environment and Soft Actor-Critic agent (stable-baselines3); financial domain, but the same simulation / reinforcement-learning loop.

EDUCATION

Engineering degree — ENSAM Meknès, IATD-SI track 2022 – 2027 (degree expected 2027)

LANGUAGES

French: working language, fluent · English: fluent written, intermediate spoken